

王锦波

2024270223

第四章作业

4-13

解：依题，时钟方程： $CP_0 = CP_1 = CP_2 = CP$

驱动方程： $J_0 = 1, J_1 = \overline{Q_0}, J_2 = \overline{Q_0} \overline{Q_1}$

$K_0 = 1, K_1 = \overline{Q_0}, K_2 = \overline{Q_0} \overline{Q_1}$

由JK触发器的特性方程：

$$Q^{n+1} = J\overline{Q}^n + \overline{K}Q^n$$

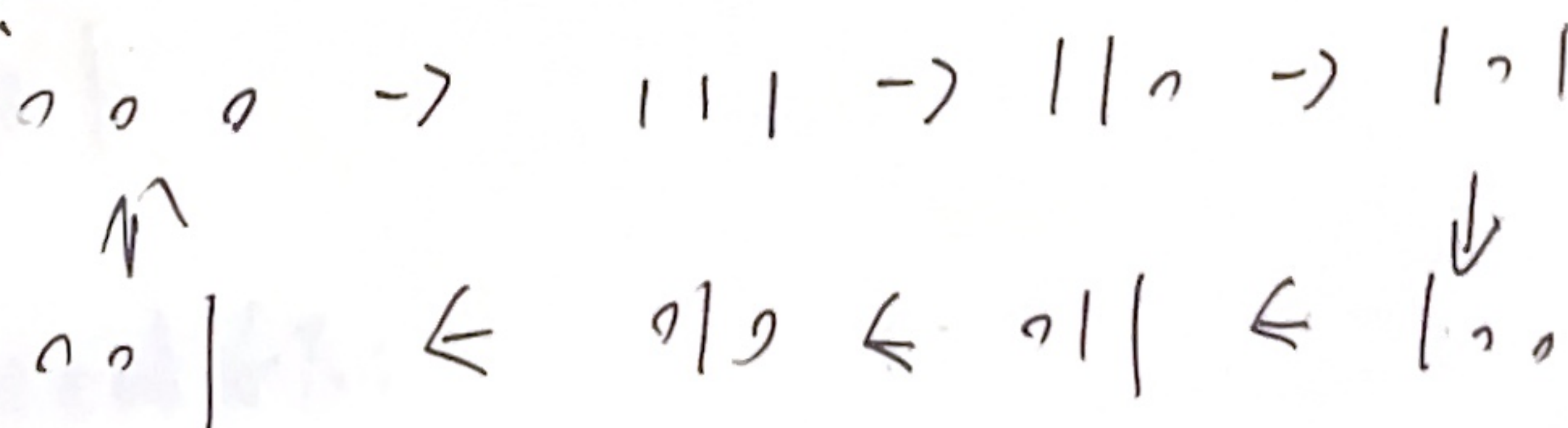
代入各触发器的状态方程：

$$Q_0^{n+1} = \overline{Q_0}^n, Q_1^{n+1} = \overline{Q_0}^n \overline{Q_1}^n + Q_0^n Q_1^n, Q_2^{n+1} = \overline{Q_0}^n \overline{Q_1}^n \overline{Q_2}^n + \overline{Q_0}^n \overline{Q_1}^n Q_2^n$$

状态转移表如下：

Q_2^n	Q_1^n	Q_0^n	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	Q_2^n	Q_1^n	Q_0^n	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}
0	0	0	1	1	1	1	0	0	0	1	1
0	0	1	0	0	0	1	0	1	1	0	0
0	1	0	0	0	1	1	1	0	1	0	1
0	1	1	0	1	0	1	1	1	1	1	0

状态图：



4-11. 解:

依题, 状态转换表如下:

Q_2^n	Q_1^n	Q_0^n	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	J_2	K_2	J_1	K_1	J_0	K_0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	0	0	0	X	1	X	1	0	X
1	1	1	X	X	X	X	X	X	X	X	X

卡诺图及化简驱动方程:

卡诺图 Q_2^n, Q_1^n, Q_0^n

$Q_2^n \backslash Q_1^n Q_0^n$	00	01	11	10
0	0	0	1	0
1	X	X	X	X

卡诺图 Q_2^n, Q_1^n, Q_0^n

$Q_2^n \backslash Q_1^n Q_0^n$	00	01	11	10
0	X	X	1	0
1	X	X	X	1

卡诺图 Q_2^n, Q_1^n, Q_0^n

$Q_2^n \backslash Q_1^n Q_0^n$	00	01	11	10
0	X	X	X	X
1	0	0	X	1

卡诺图 Q_2^n, Q_1^n, Q_0^n

$Q_2^n \backslash Q_1^n Q_0^n$	00	01	11	10
0	X	X	X	1
1	1	X	X	0

卡诺图 Q_2^n, Q_1^n, Q_0^n

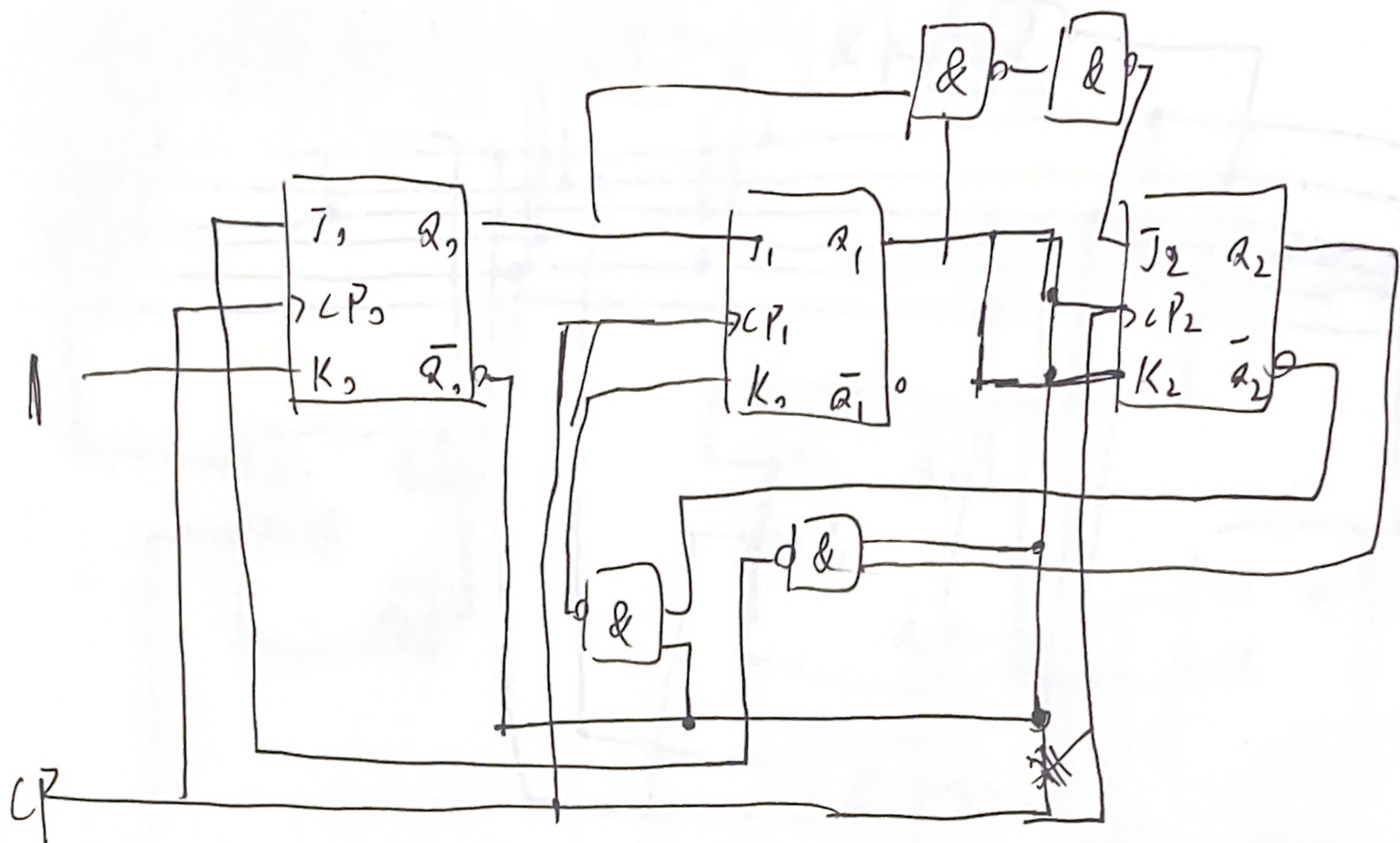
$Q_2^n \backslash Q_1^n Q_0^n$	00	01	11	10
0	0	1	X	X
1	0	1	X	X

卡诺图 Q_2^n, Q_1^n, Q_0^n

$Q_2^n \backslash Q_1^n Q_0^n$	00	01	11	10
0	X	1	1	X
1	X	1	X	X

$\Rightarrow J_2 = Q_1^n Q_0^n, K_2 = Q_1^n, J_1 = Q_0^n, K_1 = Q_0^n + Q_2^n, J_0 = \overline{Q_1^n} + \overline{Q_2^n}$
 $K_0 = 1$

设计电路如下:



4-18 解:

时钟方程: $CP_0 = CP_1 = CP_2 = CP$

状态转换表:

Q_2^n	Q_1^n	Q_0^n	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	D_2	D_1	D_0
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	0
0	1	0	0	1	1	0	1	1
0	1	1	1	0	0	1	0	0
1	0	0	1	1	1	1	1	1
1	0	1	0	0	0	0	0	0
1	1	0	1	0	1	1	0	1
1	1	1	1	1	0	1	1	0

$Q_2^n \backslash Q_1^n$	00	01	11	10
0	0	0	1	0
1	1	0	1	1

$$D_2 = Q_2^n \bar{Q}_0^n + Q_1^n Q_0^n$$

$Q_2^n \backslash Q_1^n Q_0^n$	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$D_1 = \bar{Q}_2^n \bar{Q}_1^n Q_0^n + \bar{Q}_2^n Q_1^n \bar{Q}_0^n + Q_2^n \bar{Q}_1^n Q_0^n + Q_2^n Q_1^n Q_0^n$$

$Q_2^n \backslash Q_1^n Q_0^n$	00	01	11	10
0	1	0	0	1
1	1	0	1	1

$$D_0 = Q_0^n$$

电路图如下

